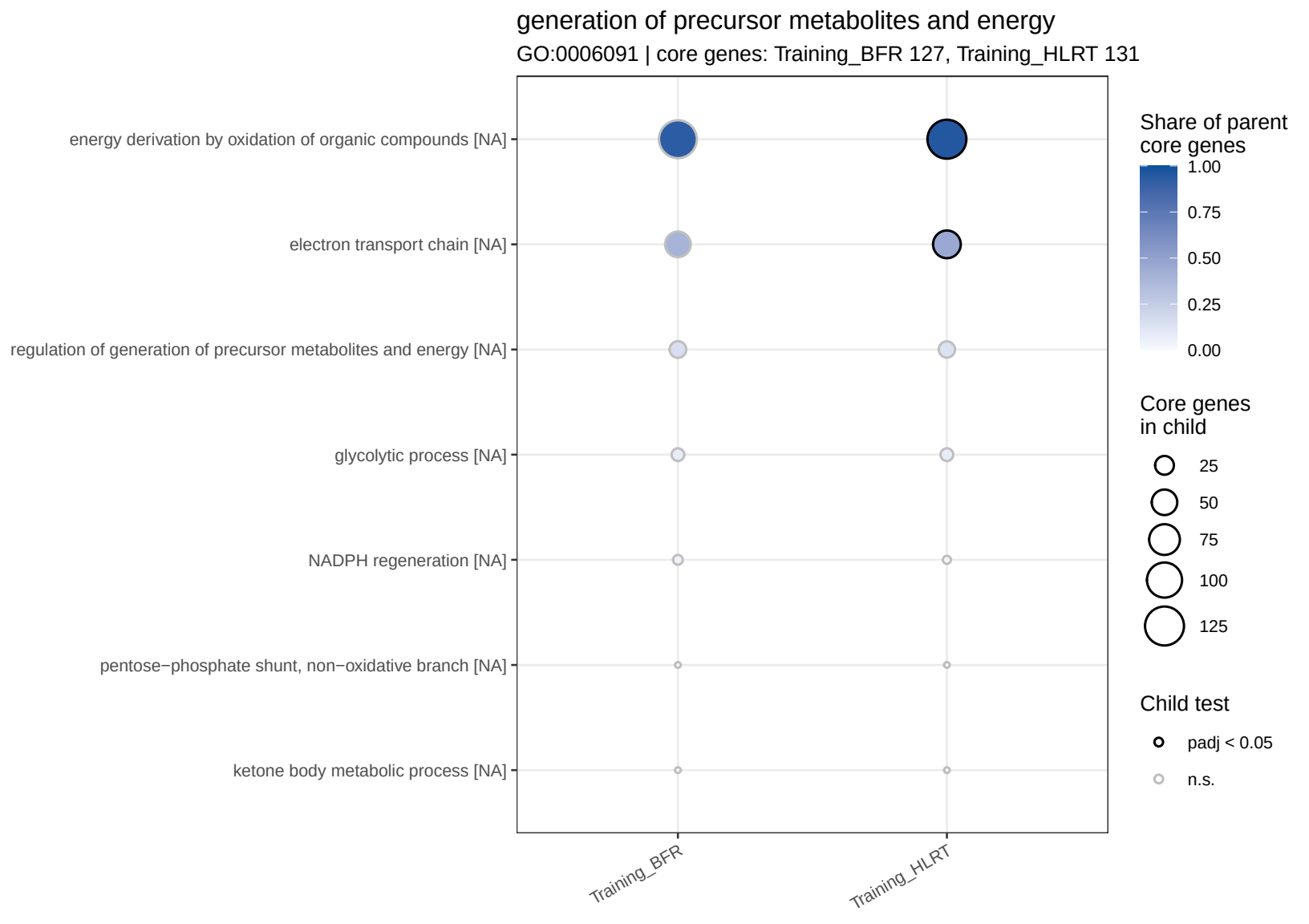
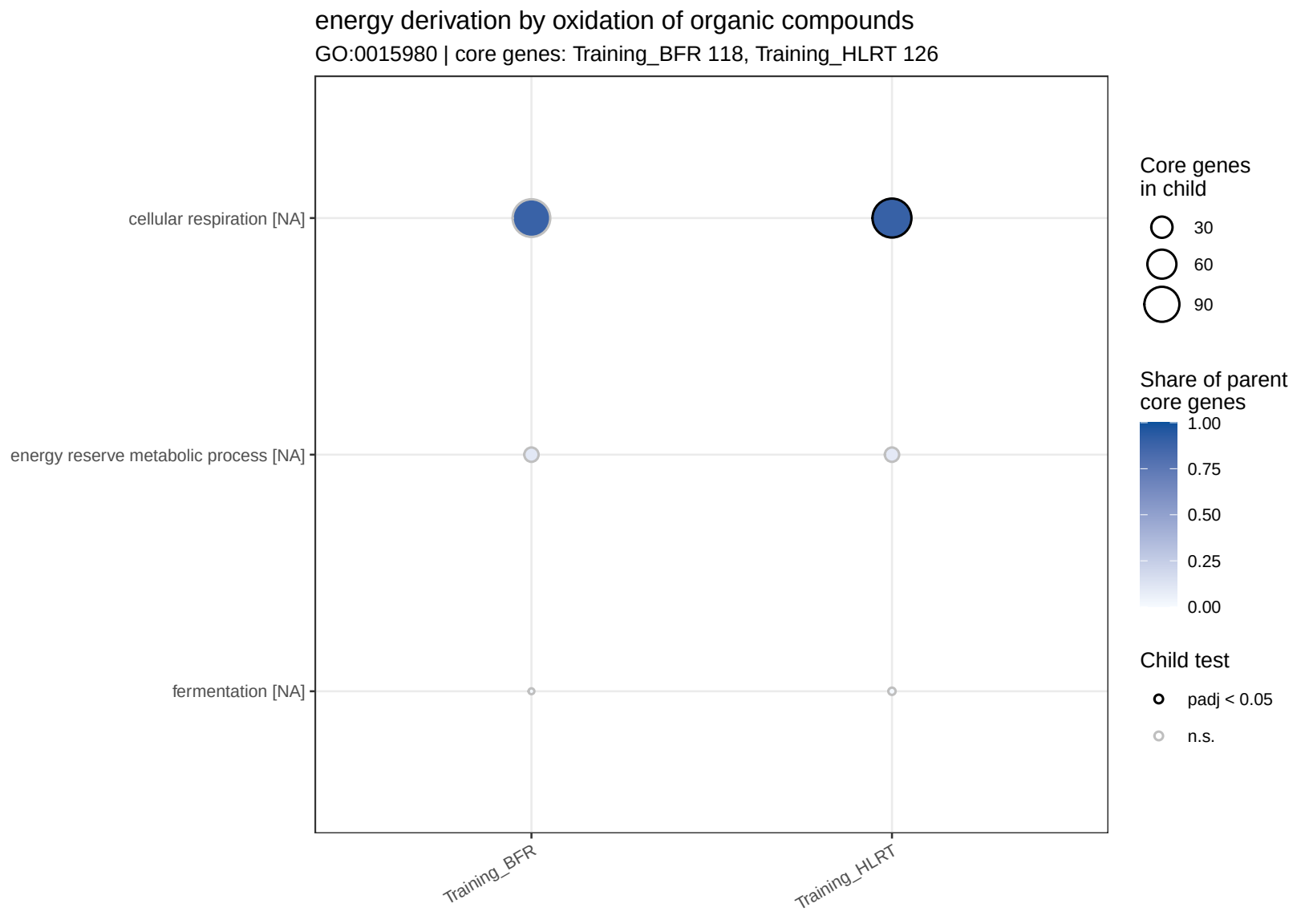
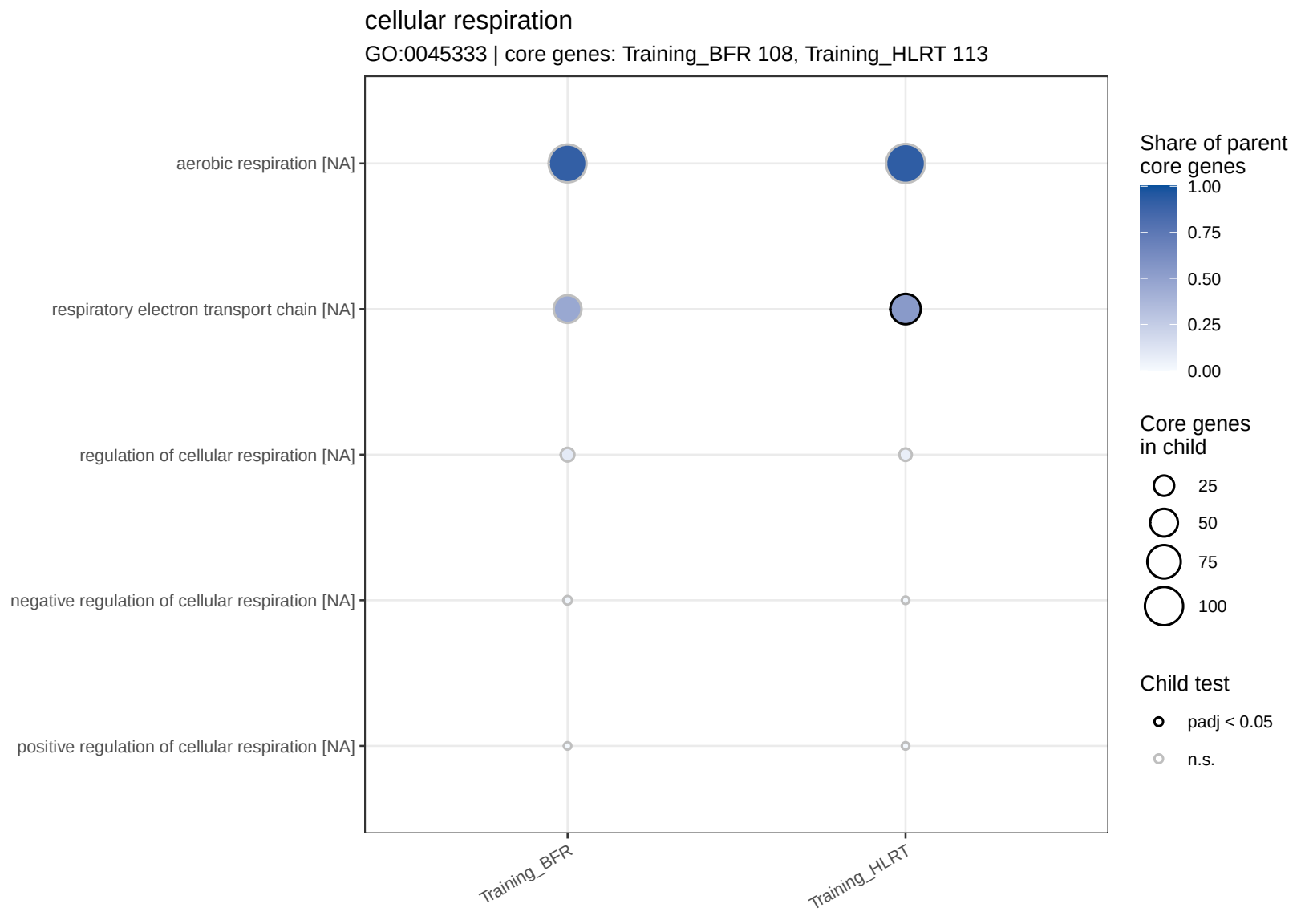
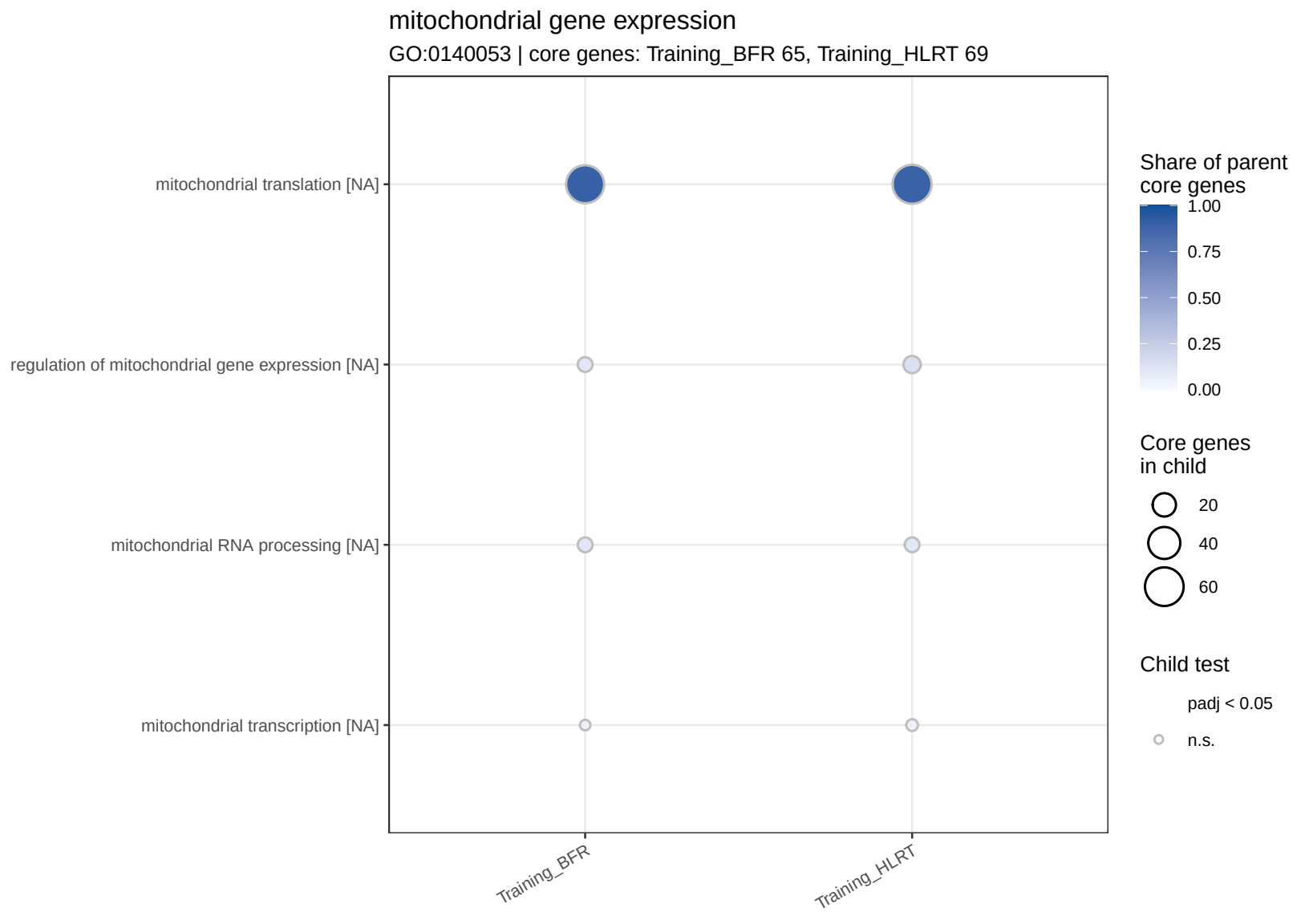
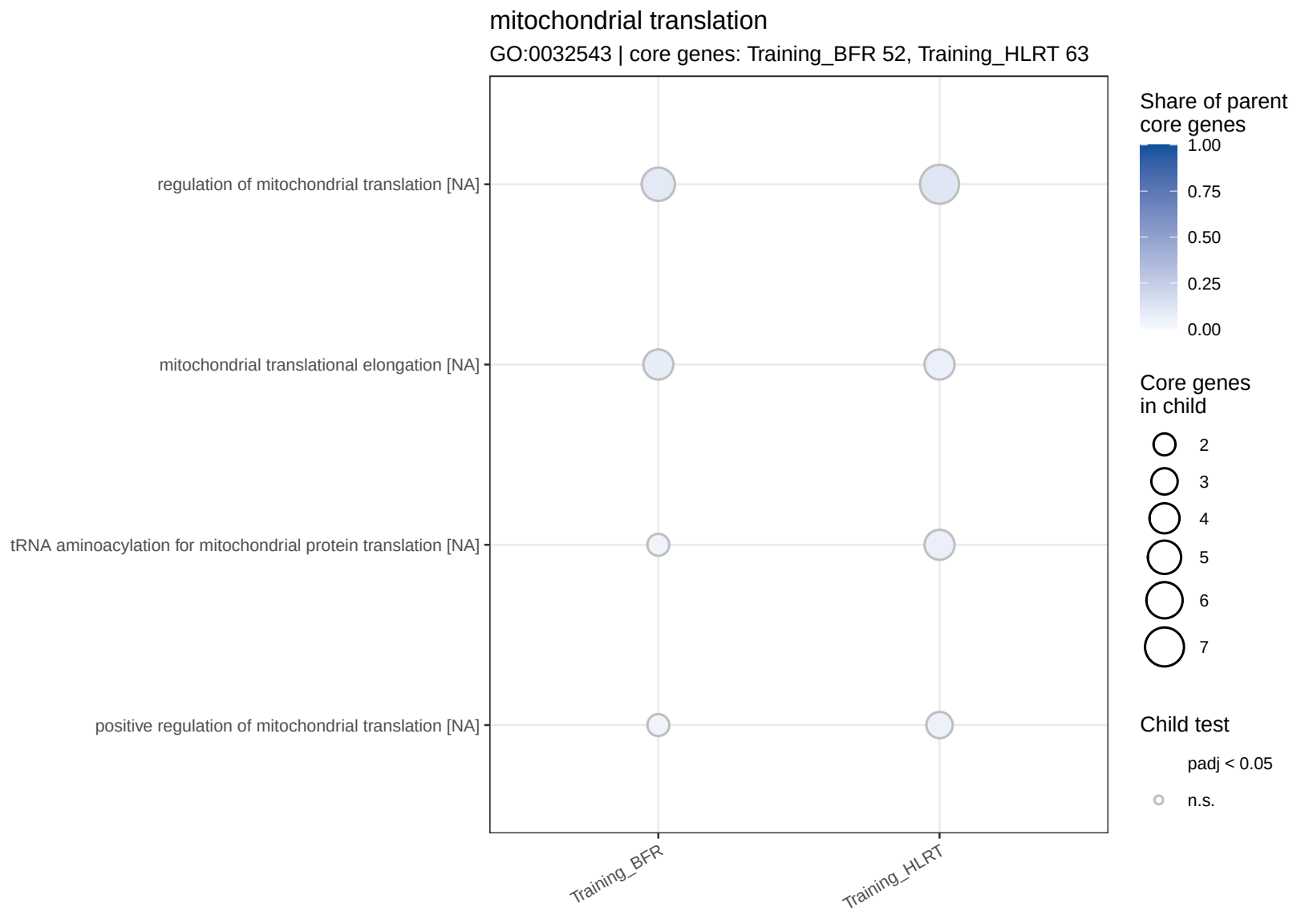
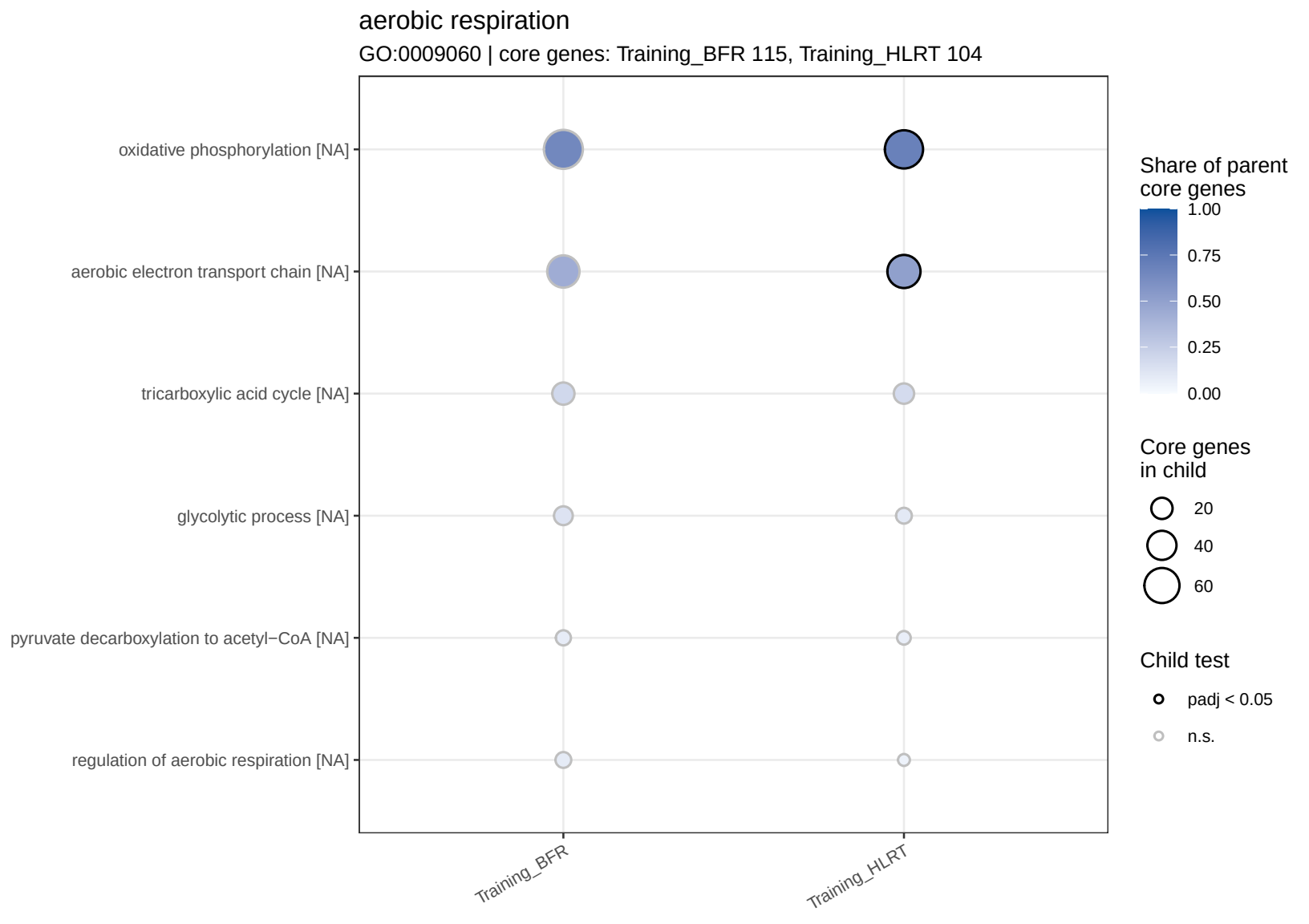

generation of precursor metabolites and energy
GO:0006091 | core genes: Training_BFR 127, Training_HLRT 131
energy derivation by oxidation of organic compounds [NA]
electron transport chain [NA]
regulation of generation of precursor metabolites and energy [NA]
glycolytic process [NA]
NADPH regeneration [NA]
pentose−phosphate shunt, non−oxidative branch [NA]
ketone body metabolic process [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
25
50
75
100
125
Child test
padj < 0.05
n.s.

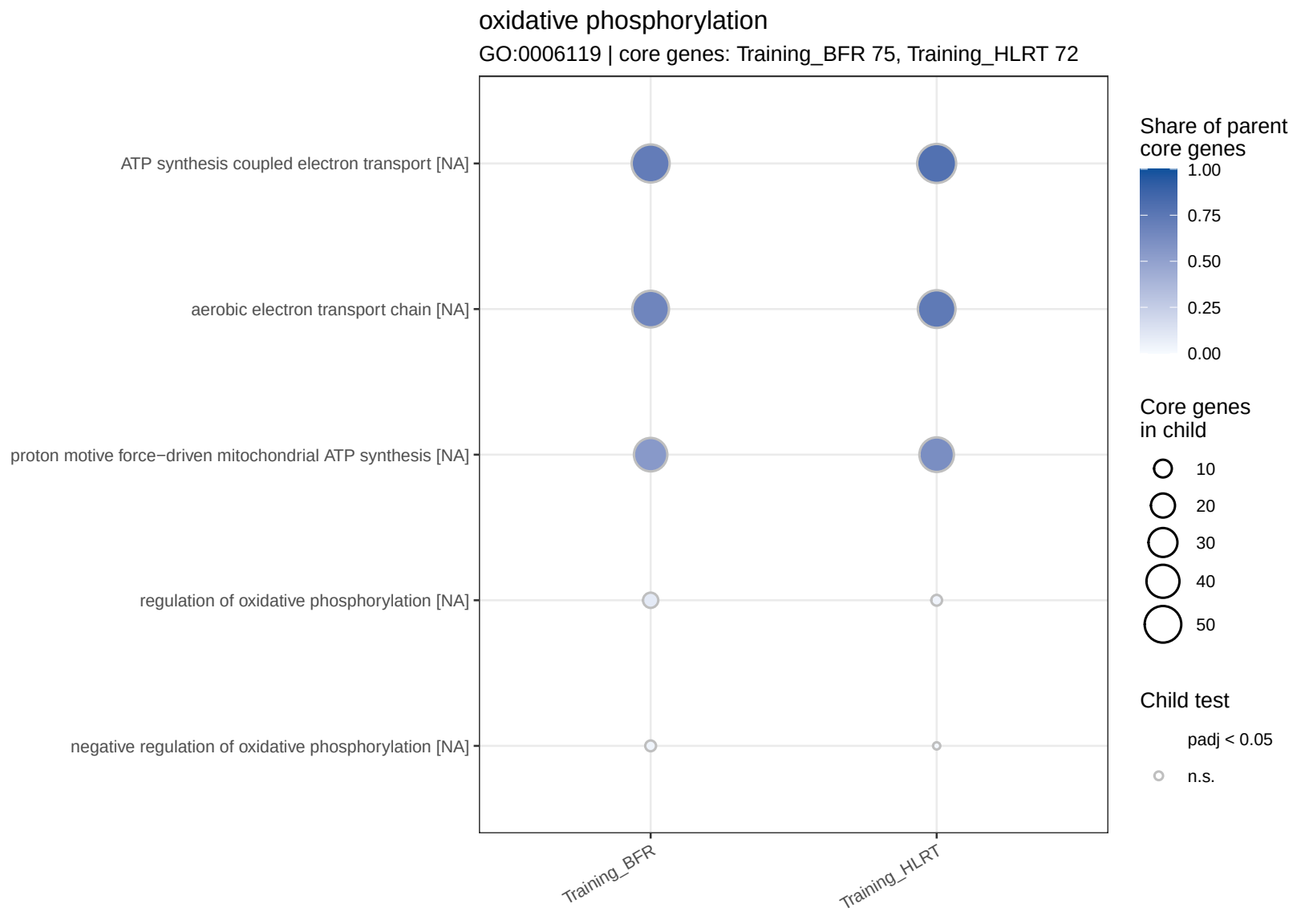

energy derivation by oxidation of organic compounds
GO:0015980 | core genes: Training_BFR 118, Training_HLRT 126
cellular respiration [NA]
energy reserve metabolic process [NA]
fermentation [NA]
Training_BFR
Training_HLRT
Core genes in child
30
60
90
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.

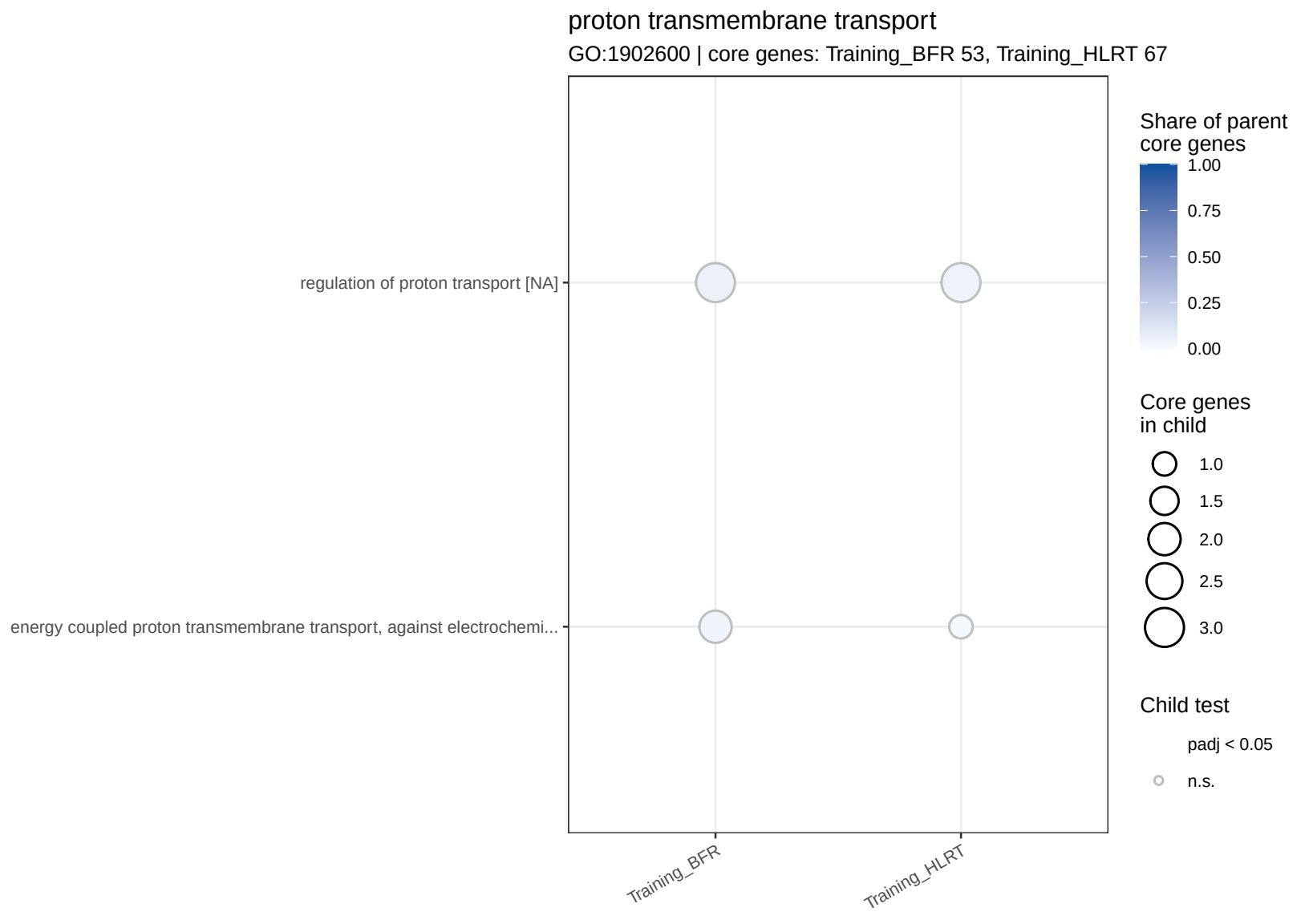

cellular respiration
GO:0045333 | core genes: Training_BFR 108, Training_HLRT 113
aerobic respiration [NA]
respiratory electron transport chain [NA]
regulation of cellular respiration [NA]
negative regulation of cellular respiration [NA]
positive regulation of cellular respiration [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
25
50
75
100
Child test
padj < 0.05
n.s.

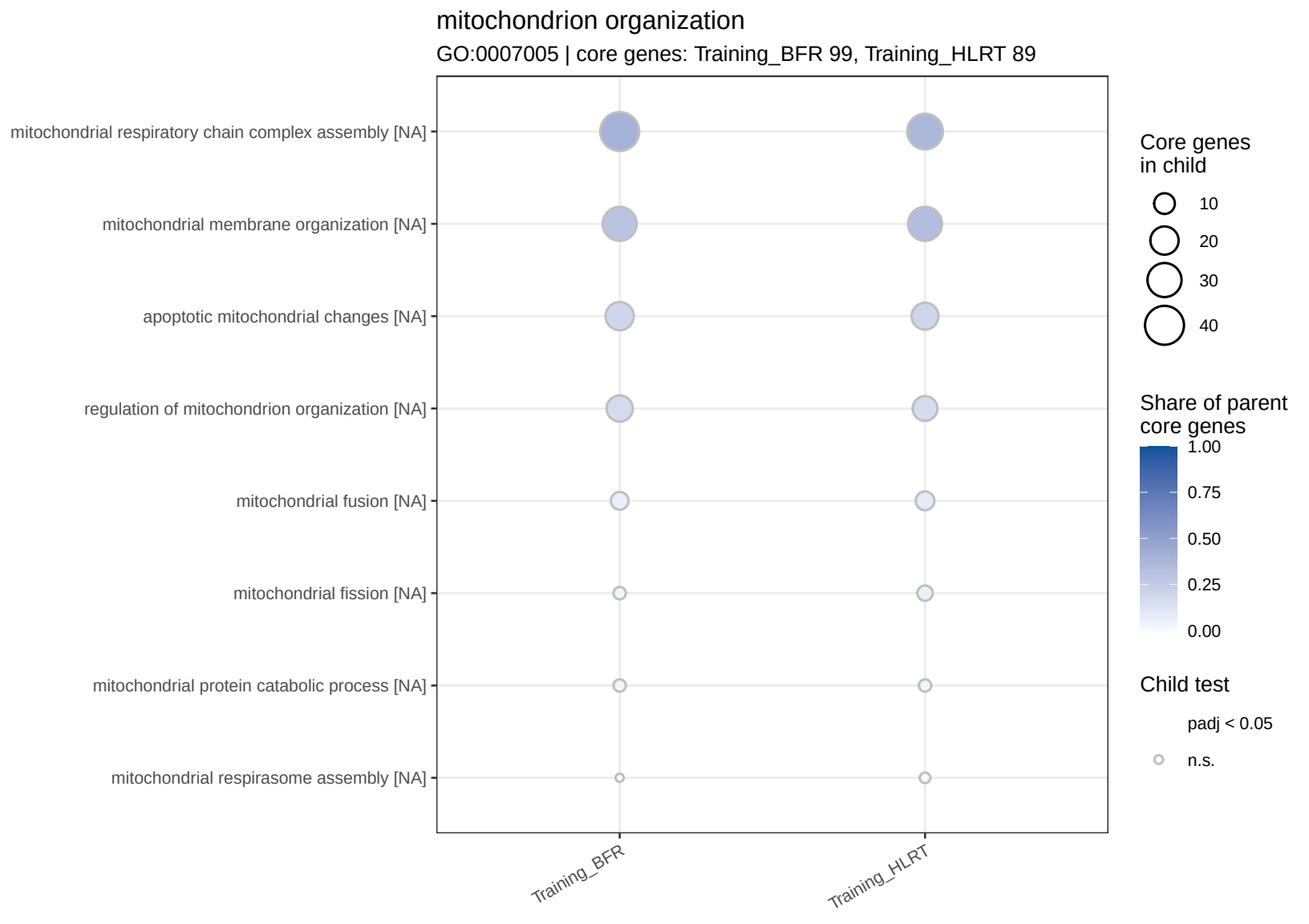

mitochondrial gene expression
GO:0140053 | core genes: Training_BFR 65, Training_HLRT 69
mitochondrial translation [NA]
regulation of mitochondrial gene expression [NA]
mitochondrial RNA processing [NA]
mitochondrial transcription [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
20
40
60
Child test
padj < 0.05
n.s.

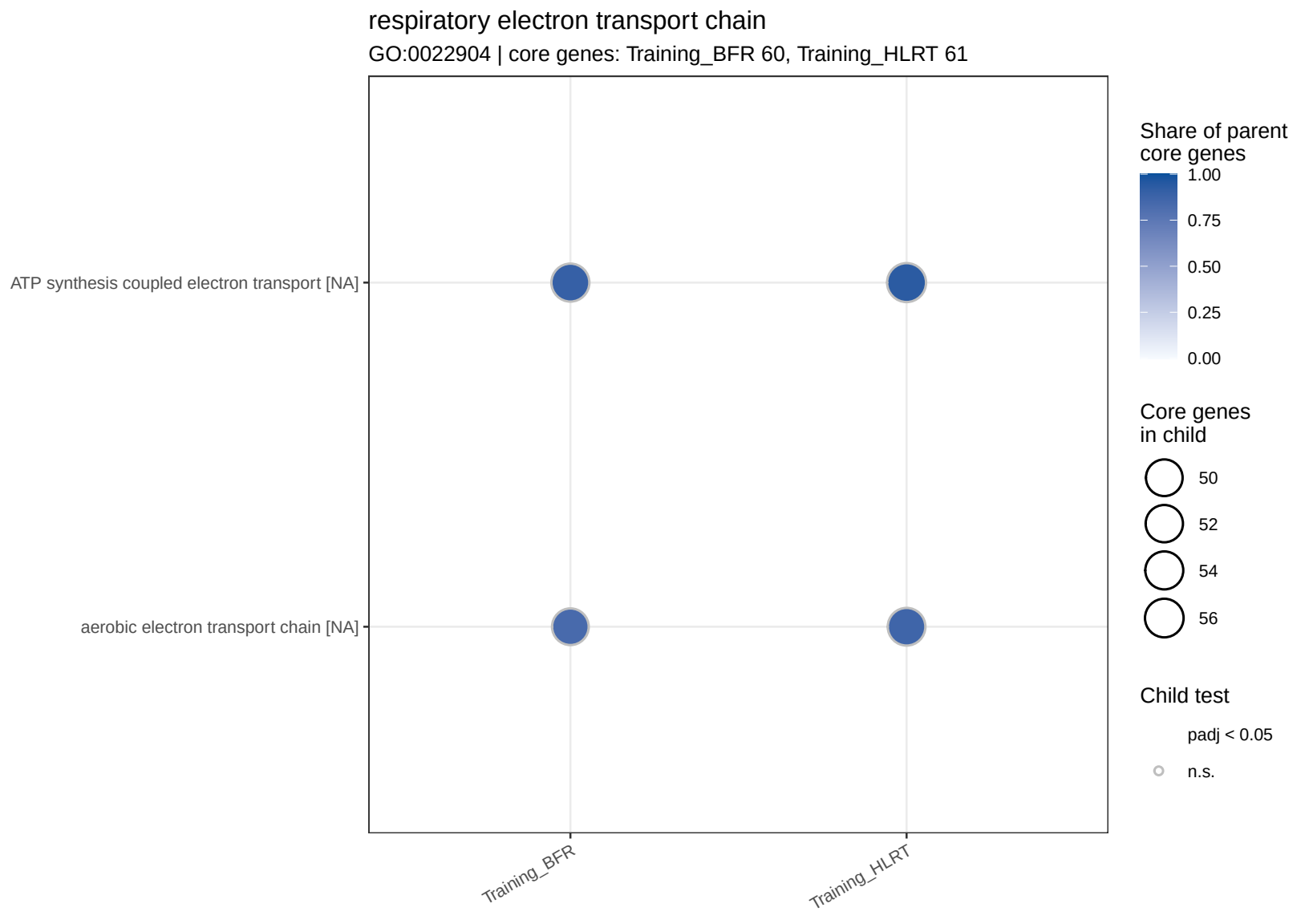

mitochondrial translation
GO:0032543 | core genes: Training_BFR 52, Training_HLRT 63
regulation of mitochondrial translation [NA]
mitochondrial translational elongation [NA]
tRNA aminoacylation for mitochondrial protein translation [NA]
positive regulation of mitochondrial translation [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
2
3
4
5
6
7
Child test
padj < 0.05
n.s.

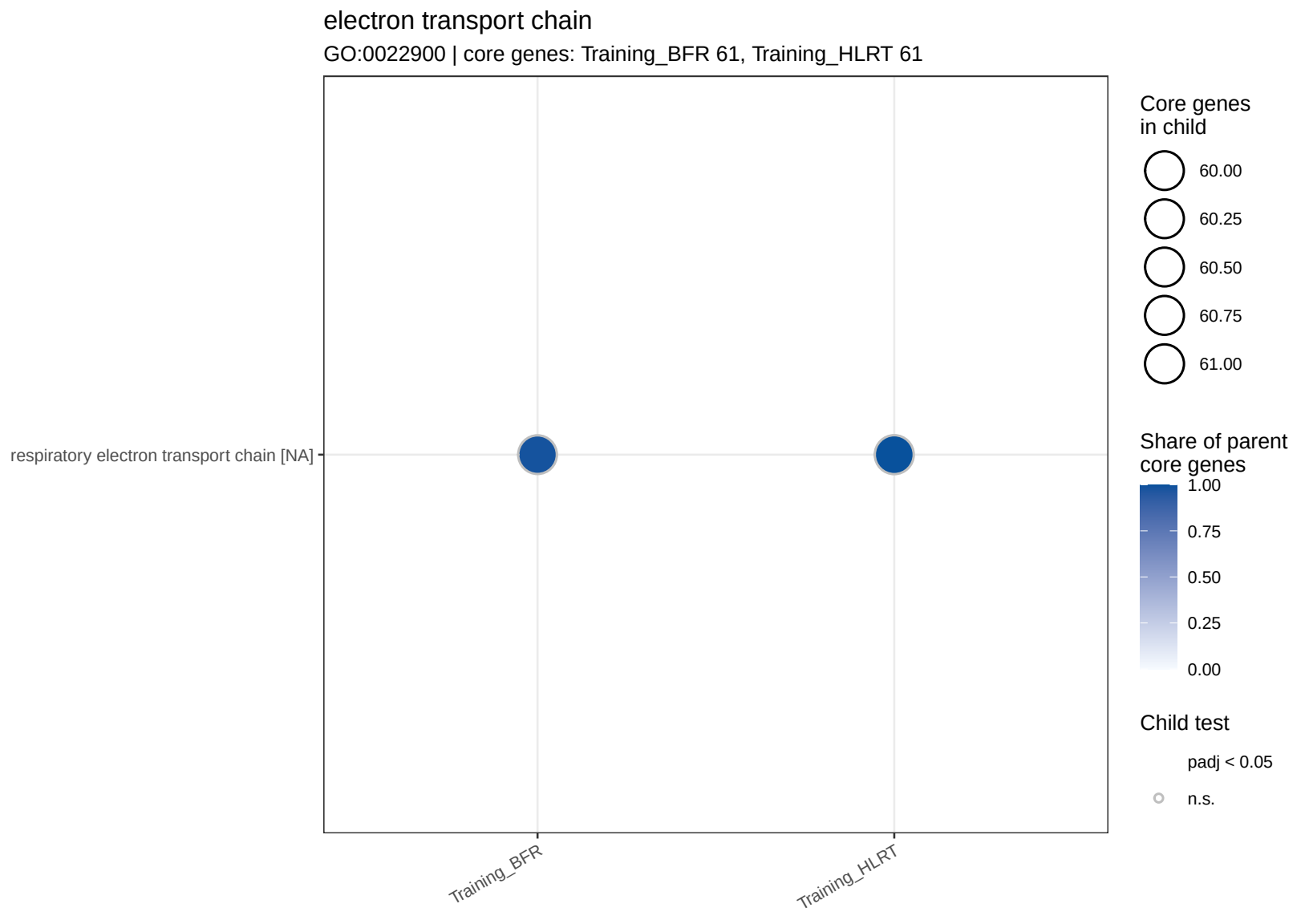

aerobic respiration
GO:0009060 | core genes: Training_BFR 115, Training_HLRT 104
oxidative phosphorylation [NA]
aerobic electron transport chain [NA]
tricarboxylic acid cycle [NA]
glycolytic process [NA]
pyruvate decarboxylation to acetyl−CoA [NA]
regulation of aerobic respiration [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
20
40
60
Child test
padj < 0.05
n.s.

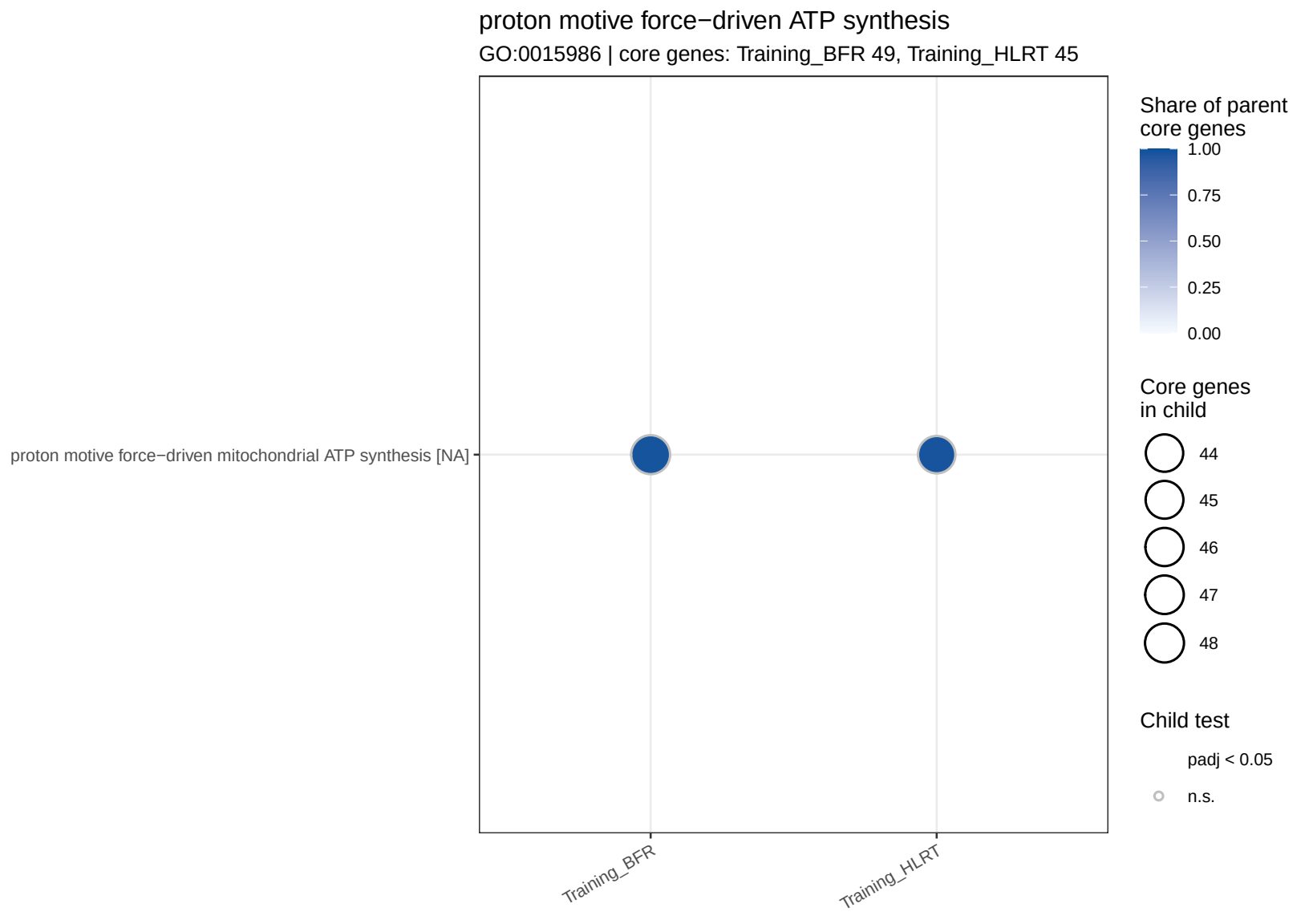

oxidative phosphorylation
GO:0006119 | core genes: Training_BFR 75, Training_HLRT 72
ATP synthesis coupled electron transport [NA]
aerobic electron transport chain [NA]
proton motive force−driven mitochondrial ATP synthesis [NA]
regulation of oxidative phosphorylation [NA]
negative regulation of oxidative phosphorylation [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
10
20
30
40
50
Child test
padj < 0.05
n.s.

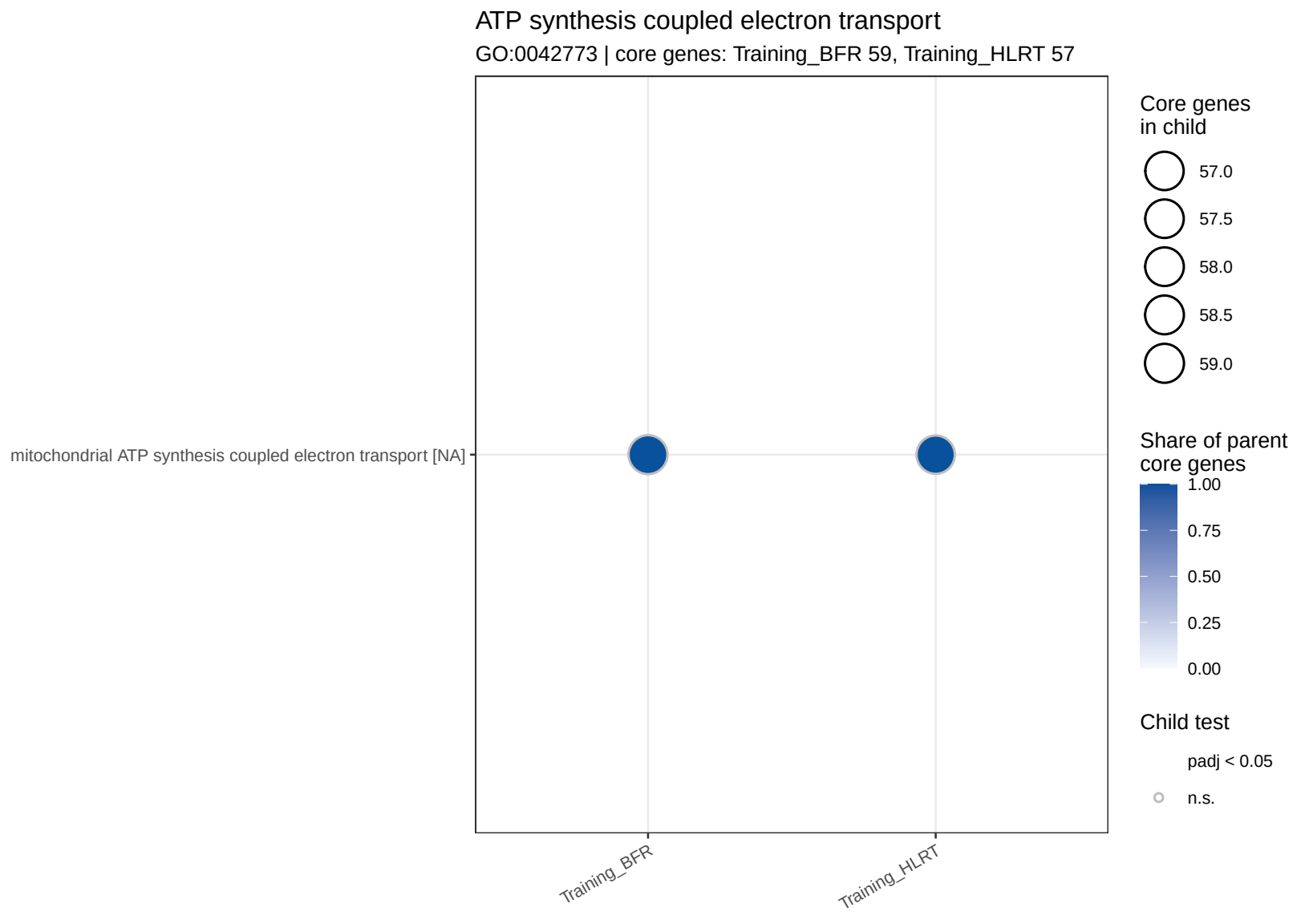

proton transmembrane transport
GO:1902600 | core genes: Training_BFR 53, Training_HLRT 67
regulation of proton transport [NA]
energy coupled proton transmembrane transport, against electrochemi...
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
1.0
1.5
2.0
2.5
3.0
Child test
padj < 0.05
n.s.

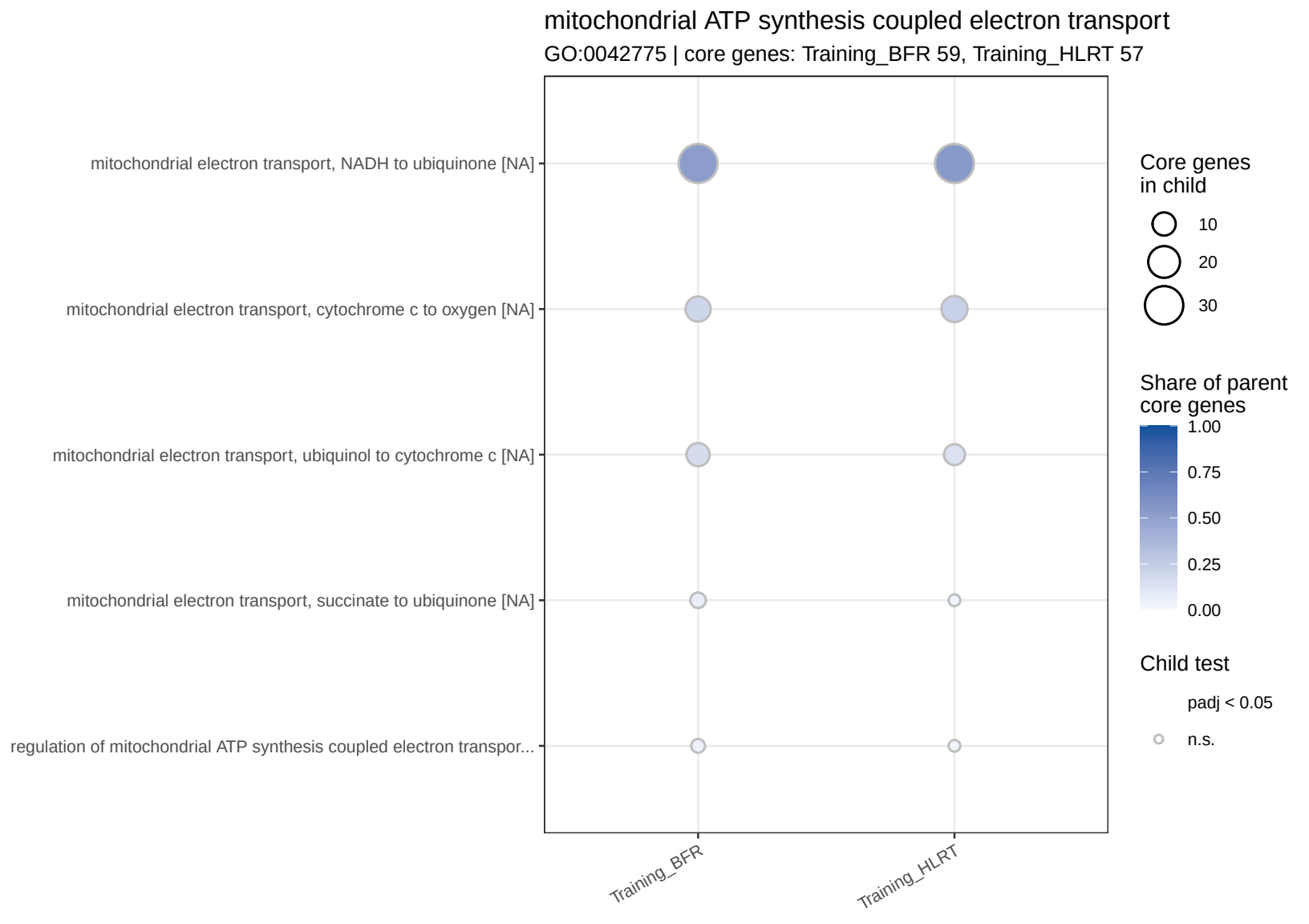

mitochondrion organization
GO:0007005 | core genes: Training_BFR 99, Training_HLRT 89
mitochondrial respiratory chain complex assembly [NA]
mitochondrial membrane organization [NA]
apoptotic mitochondrial changes [NA]
regulation of mitochondrion organization [NA]
mitochondrial fusion [NA]
mitochondrial fission [NA]
mitochondrial protein catabolic process [NA]
mitochondrial respirasome assembly [NA]
Training_BFR
Training_HLRT
Core genes in child
10
20
30
40
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.

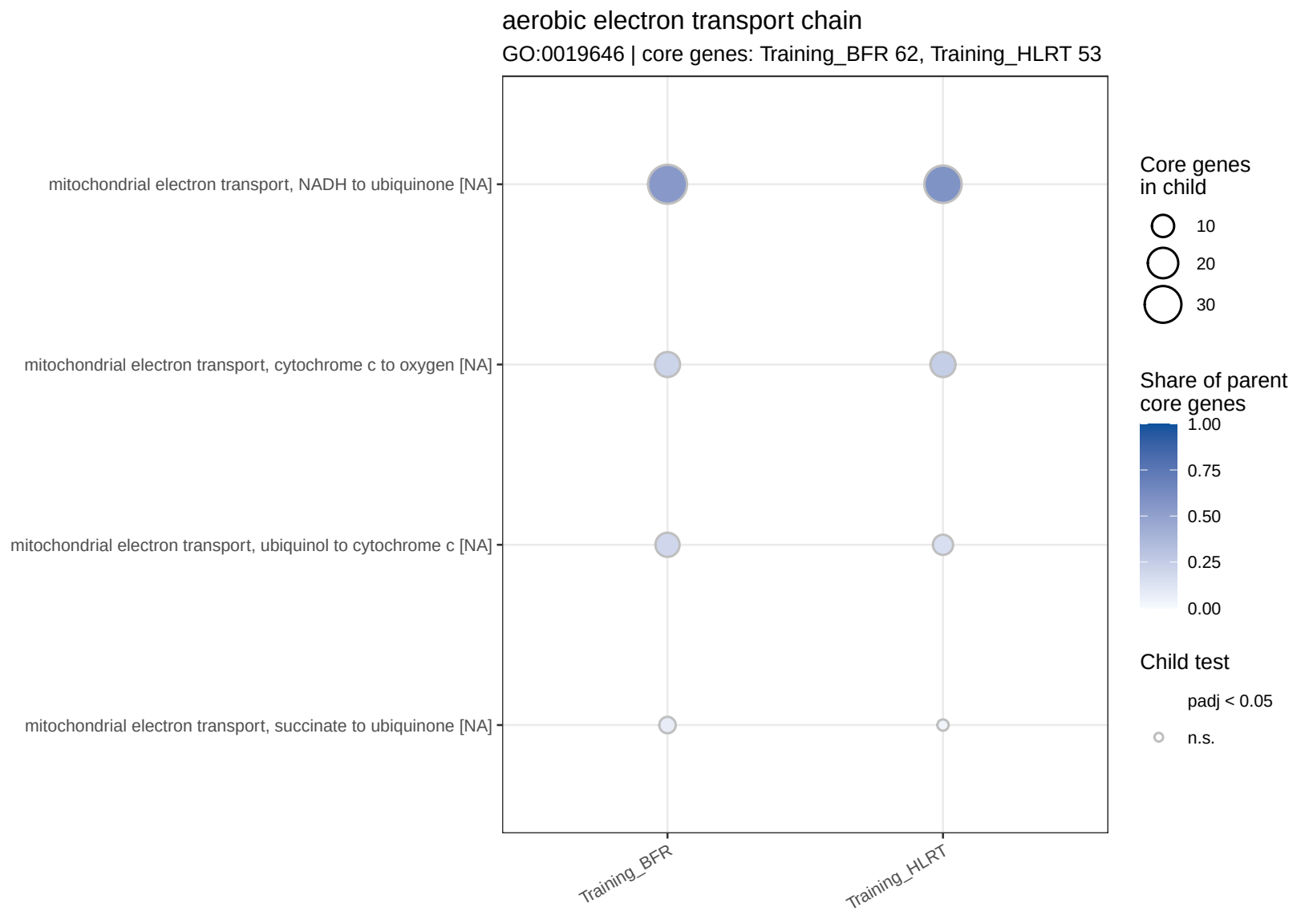

respiratory electron transport chain
GO:0022904 | core genes: Training_BFR 60, Training_HLRT 61
ATP synthesis coupled electron transport [NA]
aerobic electron transport chain [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
50
52
54
56
Child test
padj < 0.05
n.s.

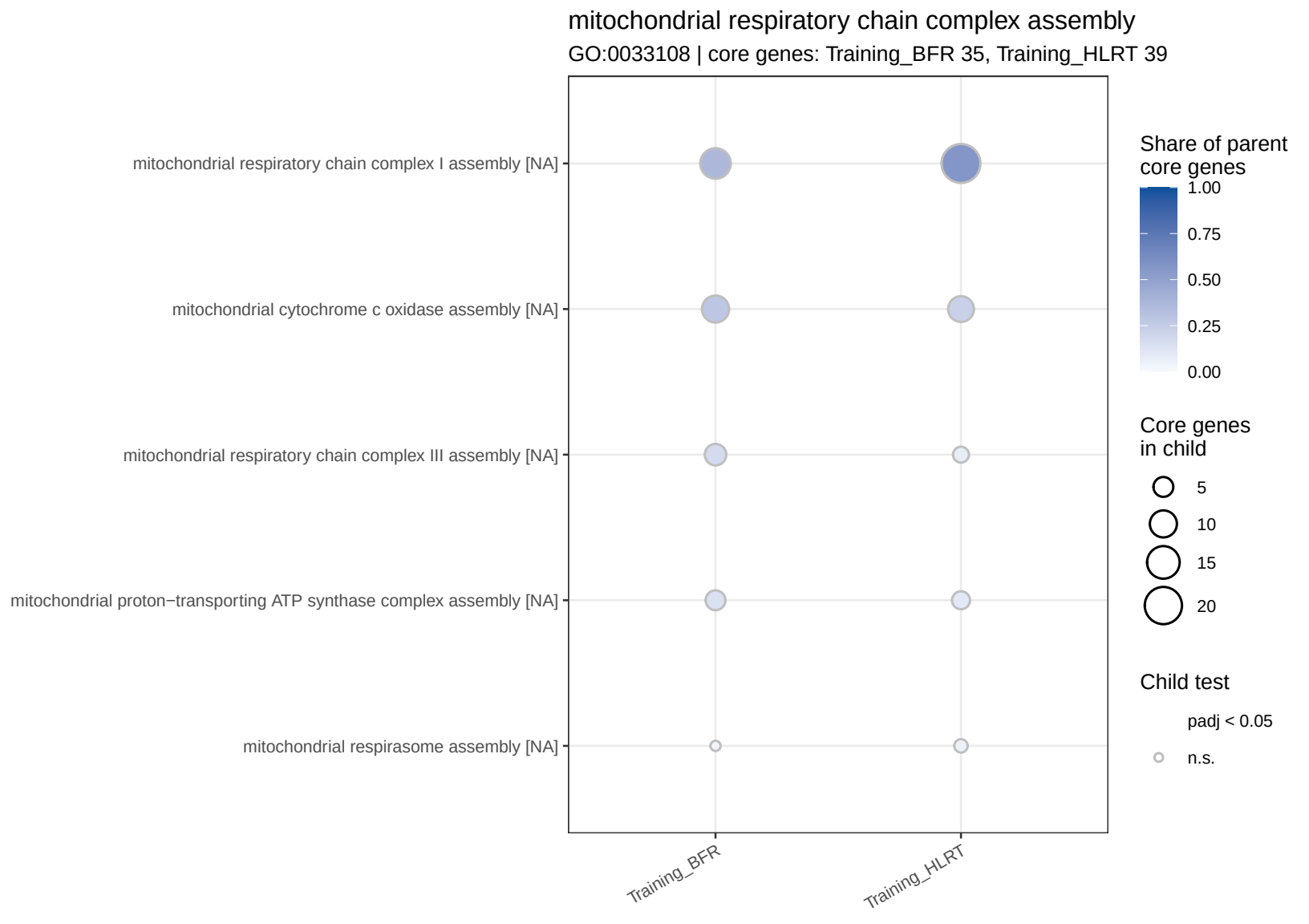

electron transport chain
GO:0022900 | core genes: Training_BFR 61, Training_HLRT 61
respiratory electron transport chain [NA]
Training_BFR
Training_HLRT
Core genes in child
60.00
60.25
60.50
60.75
61.00
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.

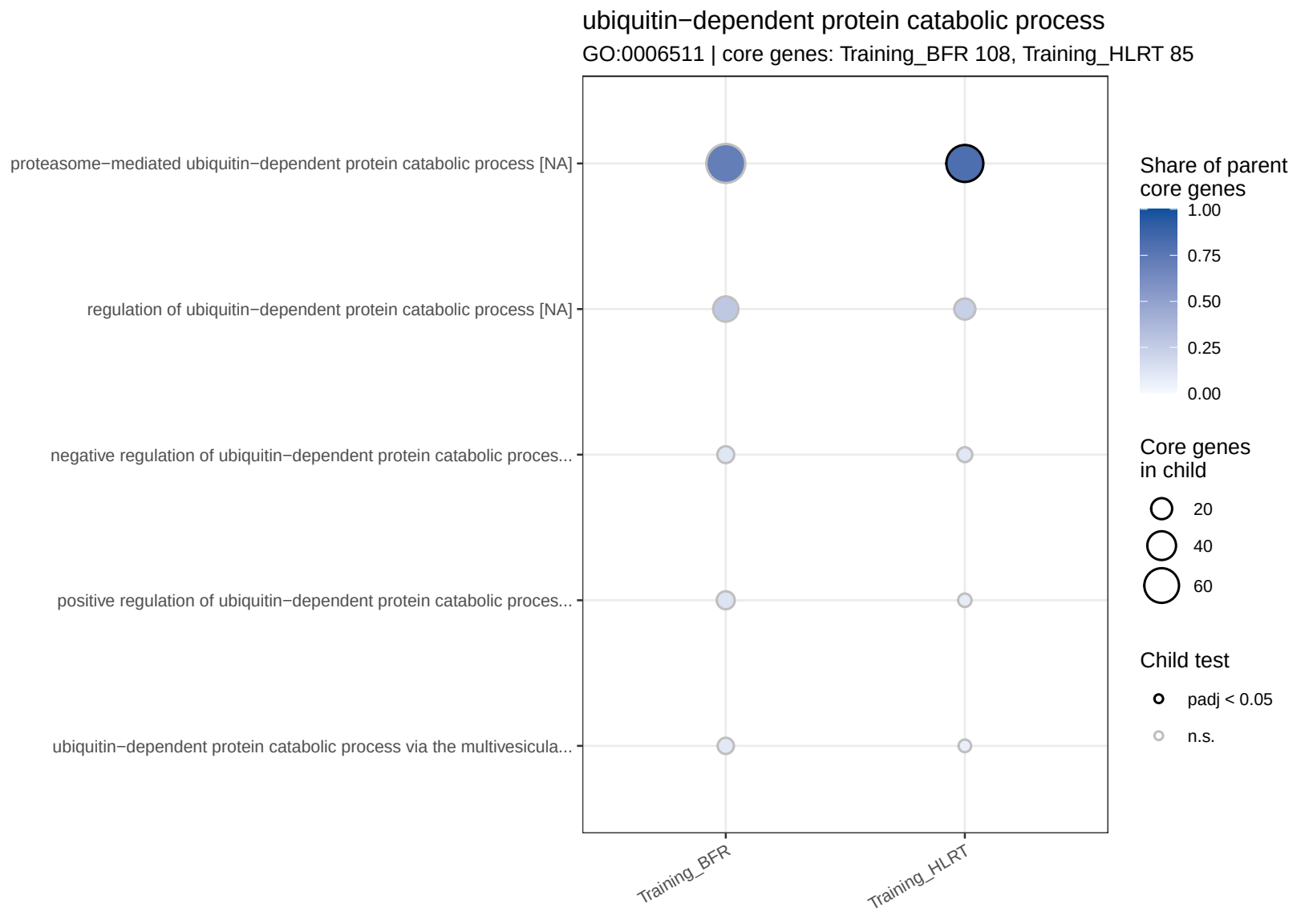

proton motive force−driven ATP synthesis
GO:0015986 | core genes: Training_BFR 49, Training_HLRT 45
proton motive force−driven mitochondrial ATP synthesis [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
44
45
46
47
48
Child test
padj < 0.05
n.s.

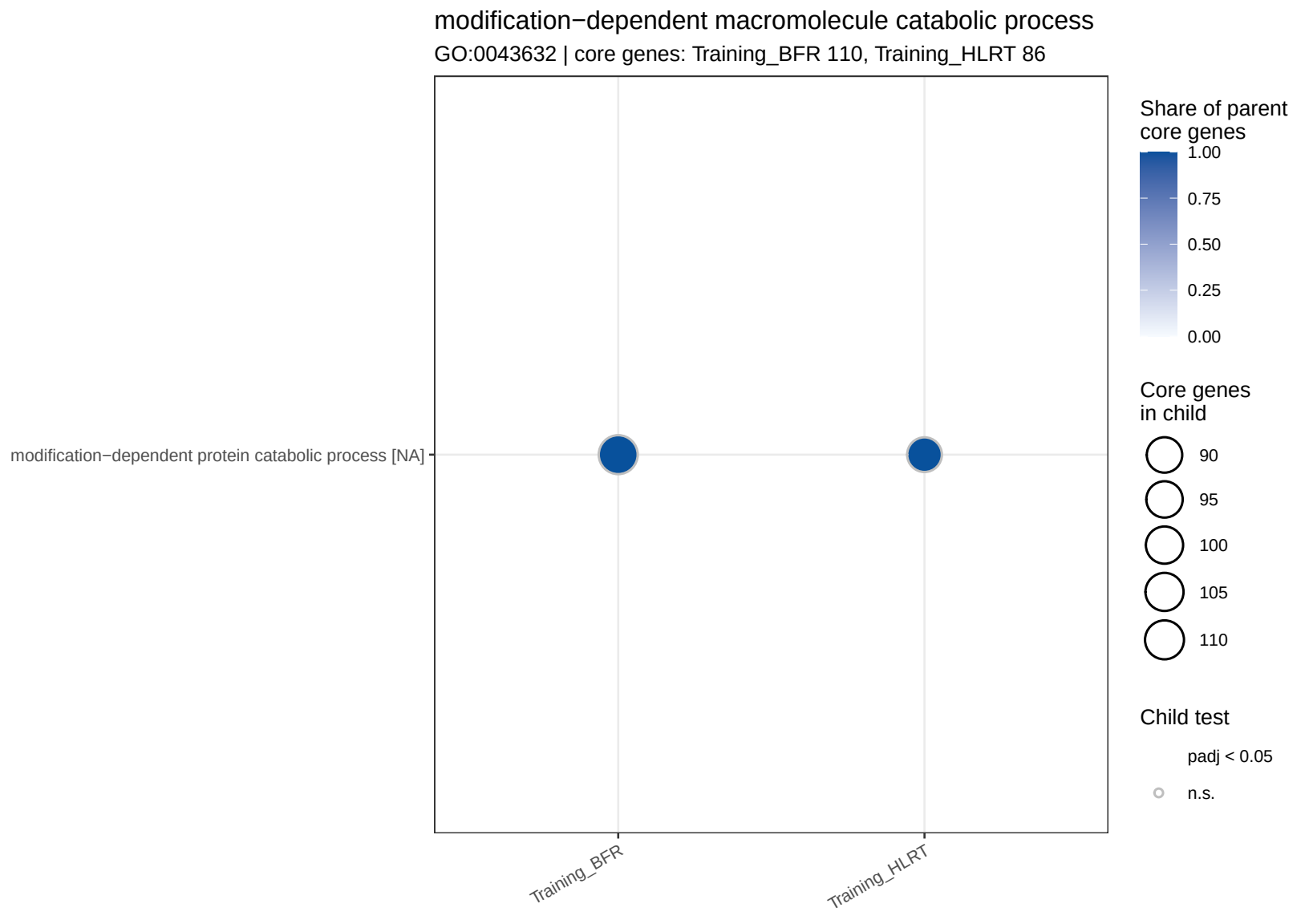

ATP synthesis coupled electron transport
GO:0042773 | core genes: Training_BFR 59, Training_HLRT 57
mitochondrial ATP synthesis coupled electron transport [NA]
Training_BFR
Training_HLRT
Core genes in child
57.0
57.5
58.0
58.5
59.0
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.

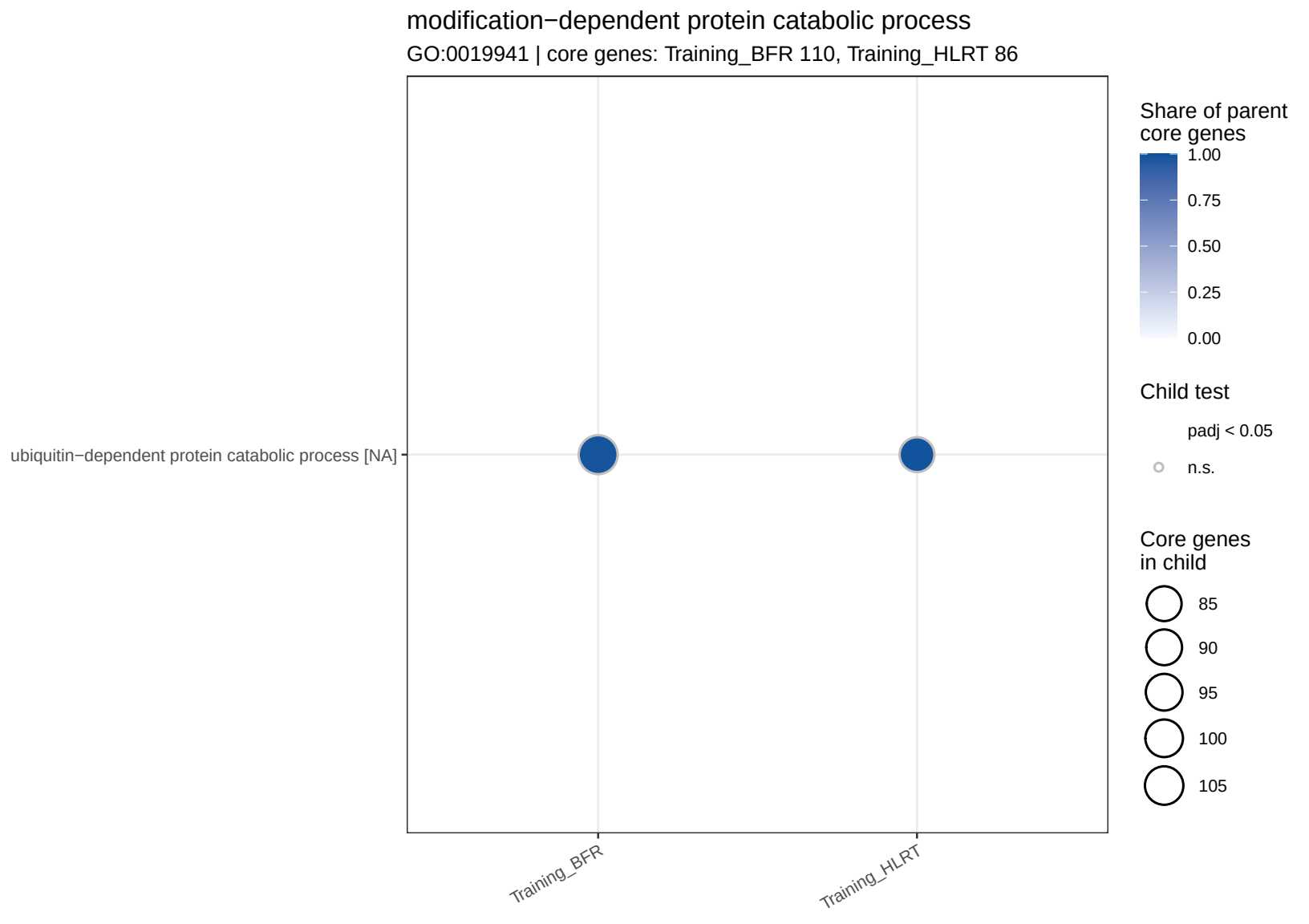

mitochondrial ATP synthesis coupled electron transport
GO:0042775 | core genes: Training_BFR 59, Training_HLRT 57
mitochondrial electron transport, NADH to ubiquinone [NA]
mitochondrial electron transport, cytochrome c to oxygen [NA]
mitochondrial electron transport, ubiquinol to cytochrome c [NA]
mitochondrial electron transport, succinate to ubiquinone [NA]
regulation of mitochondrial ATP synthesis coupled electron transpor...
Training_BFR
Training_HLRT
Core genes in child
10
20
30
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.

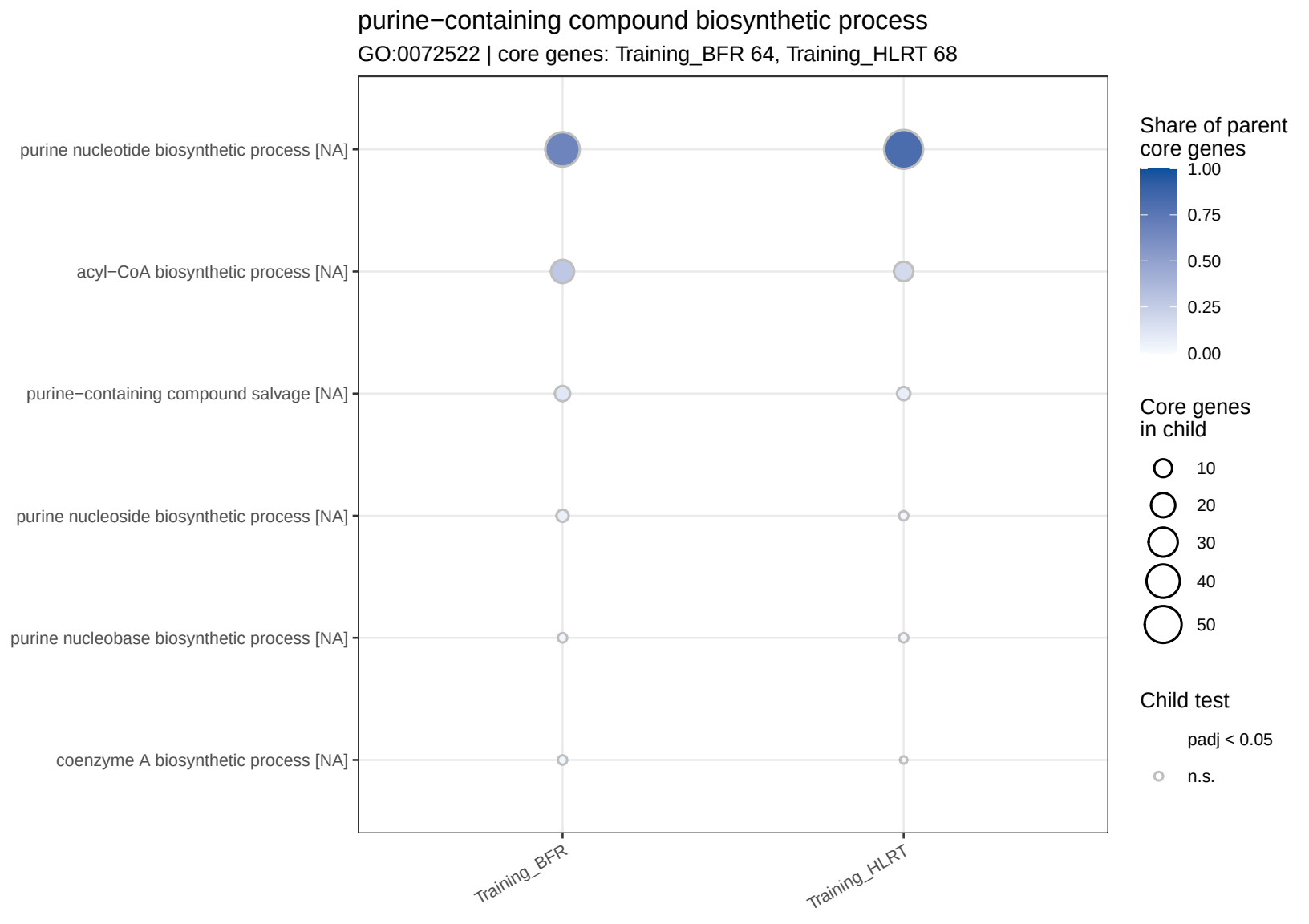

aerobic electron transport chain
GO:0019646 | core genes: Training_BFR 62, Training_HLRT 53
mitochondrial electron transport, NADH to ubiquinone [NA]
mitochondrial electron transport, cytochrome c to oxygen [NA]
mitochondrial electron transport, ubiquinol to cytochrome c [NA]
mitochondrial electron transport, succinate to ubiquinone [NA]
Training_BFR
Training_HLRT
Core genes in child
10
20
30
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.

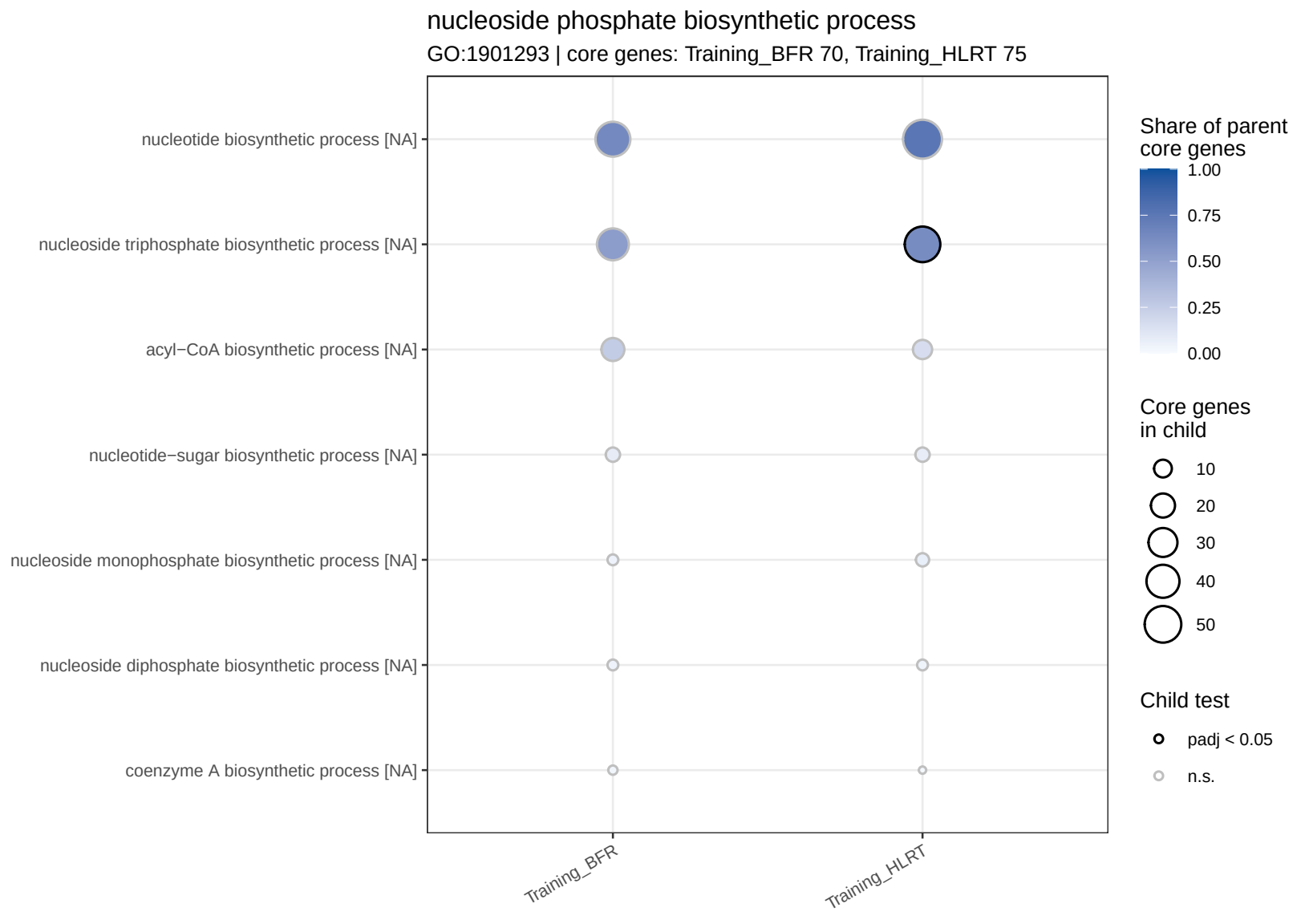

mitochondrial respiratory chain complex assembly
GO:0033108 | core genes: Training_BFR 35, Training_HLRT 39
mitochondrial respiratory chain complex I assembly [NA]
mitochondrial cytochrome c oxidase assembly [NA]
mitochondrial respiratory chain complex III assembly [NA]
mitochondrial proton−transporting ATP synthase complex assembly [NA]
mitochondrial respirasome assembly [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
5
10
15
20
Child test
padj < 0.05
n.s.

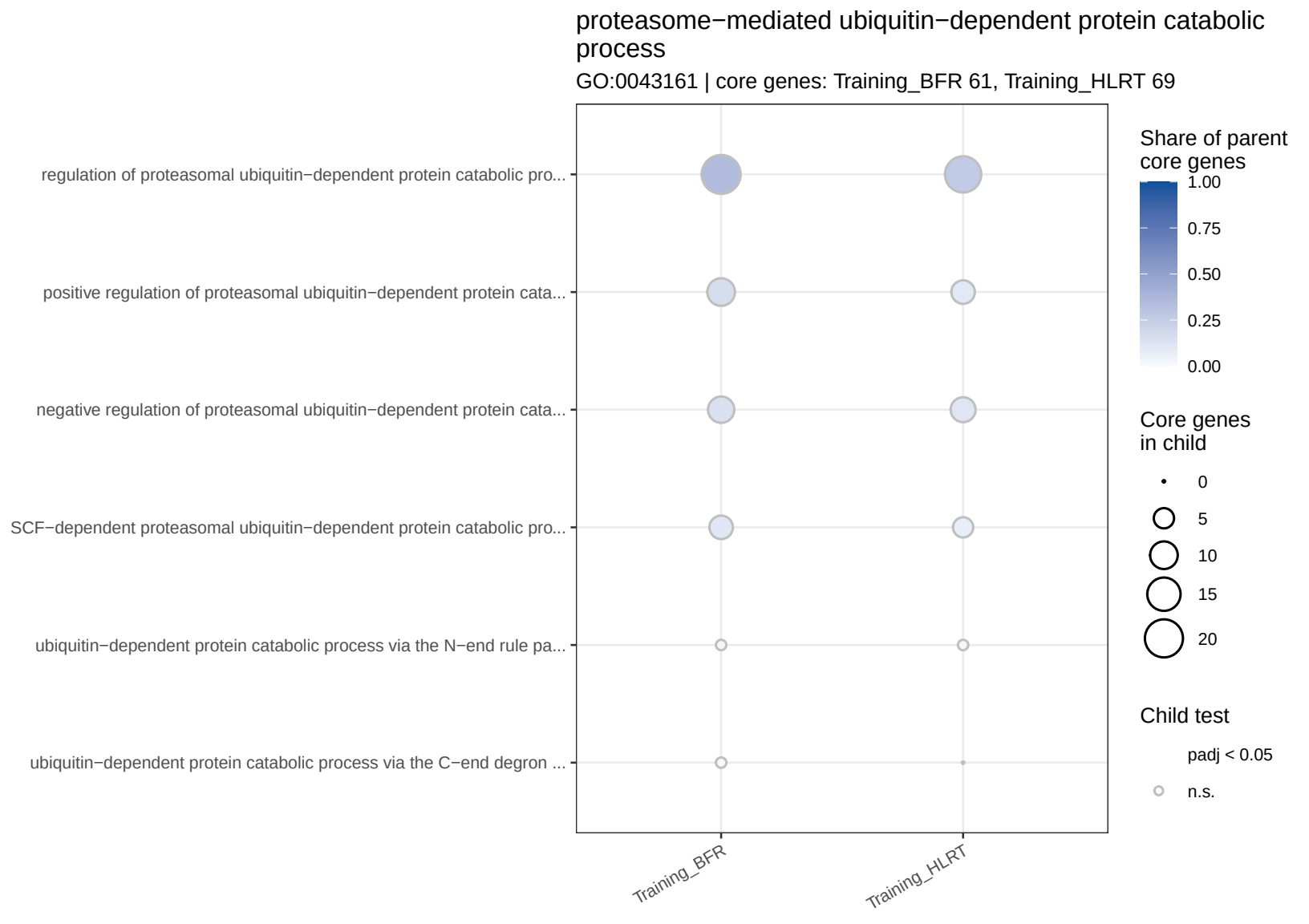

ubiquitin−dependent protein catabolic process
GO:0006511 | core genes: Training_BFR 108, Training_HLRT 85
proteasome−mediated ubiquitin−dependent protein catabolic process [NA]
regulation of ubiquitin−dependent protein catabolic process [NA]
negative regulation of ubiquitin−dependent protein catabolic proces...
positive regulation of ubiquitin−dependent protein catabolic proces...
ubiquitin−dependent protein catabolic process via the multivesicula...
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
20
40
60
Child test
padj < 0.05
n.s.

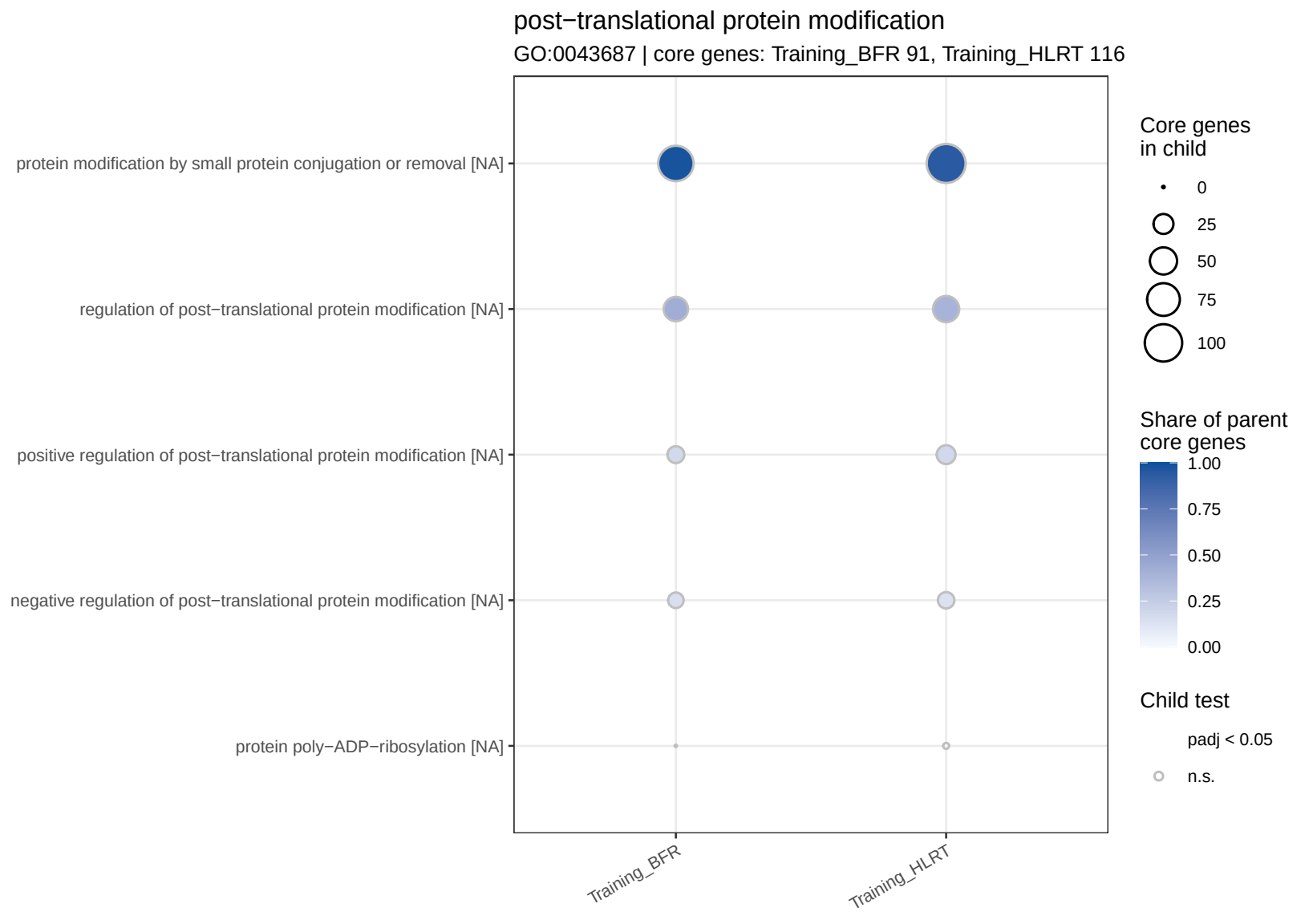

modification−dependent macromolecule catabolic process
GO:0043632 | core genes: Training_BFR 110, Training_HLRT 86
modification−dependent protein catabolic process [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
90
95
100
105
110
Child test
padj < 0.05
n.s.

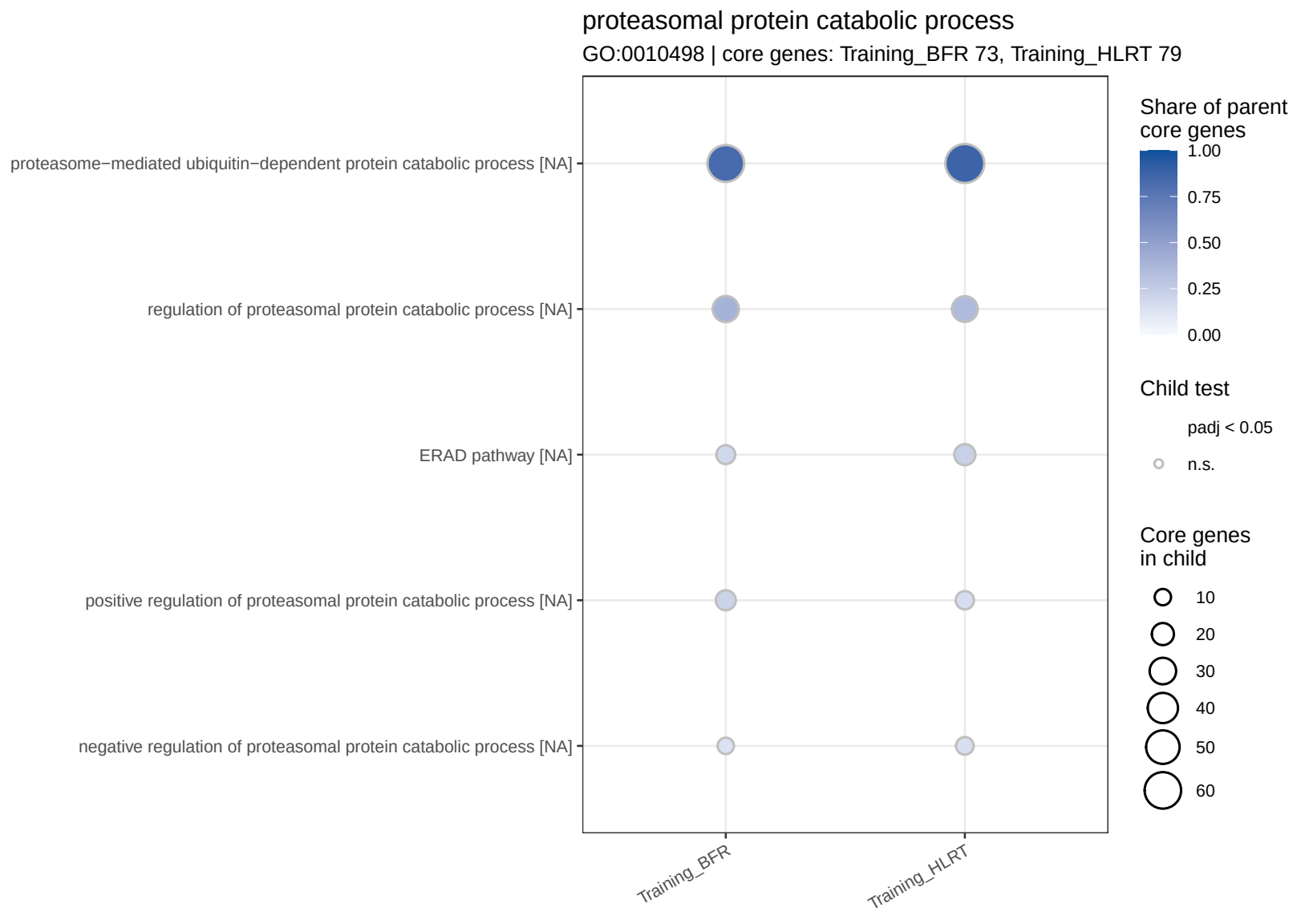

modification−dependent protein catabolic process
GO:0019941 | core genes: Training_BFR 110, Training_HLRT 86
ubiquitin−dependent protein catabolic process [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.
Core genes in child
85
90
95
100
105

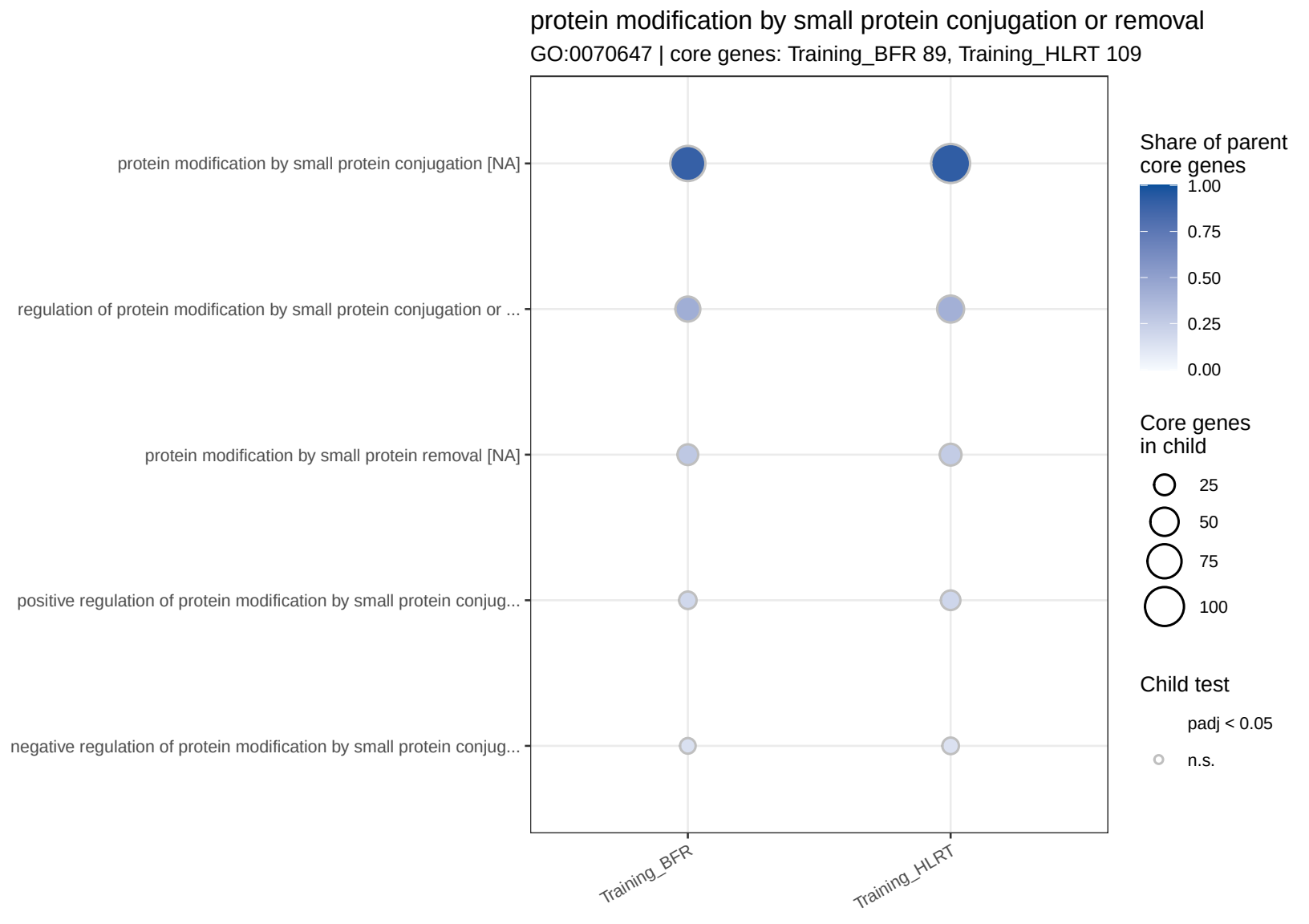

purine−containing compound biosynthetic process
GO:0072522 | core genes: Training_BFR 64, Training_HLRT 68
purine nucleotide biosynthetic process [NA]
acyl−CoA biosynthetic process [NA]
purine−containing compound salvage [NA]
purine nucleoside biosynthetic process [NA]
purine nucleobase biosynthetic process [NA]
coenzyme A biosynthetic process [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
10
20
30
40
50
Child test
padj < 0.05
n.s.


nucleoside phosphate biosynthetic process
GO:1901293 | core genes: Training_BFR 70, Training_HLRT 75
nucleotide biosynthetic process [NA]
nucleoside triphosphate biosynthetic process [NA]
acyl−CoA biosynthetic process [NA]
nucleotide−sugar biosynthetic process [NA]
nucleoside monophosphate biosynthetic process [NA]
nucleoside diphosphate biosynthetic process [NA]
coenzyme A biosynthetic process [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
10
20
30
40
50
Child test
padj < 0.05
n.s.


proteasome−mediated ubiquitin−dependent protein catabolic process
GO:0043161 | core genes: Training_BFR 61, Training_HLRT 69
regulation of proteasomal ubiquitin−dependent protein catabolic pro...
positive regulation of proteasomal ubiquitin−dependent protein cata...
negative regulation of proteasomal ubiquitin−dependent protein cata...
SCF−dependent proteasomal ubiquitin−dependent protein catabolic pro...
ubiquitin−dependent protein catabolic process via the N−end rule pa...
ubiquitin−dependent protein catabolic process via the C−end degron ...
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
0
5
10
15
20
Child test
padj < 0.05
n.s.


post−translational protein modification
GO:0043687 | core genes: Training_BFR 91, Training_HLRT 116
protein modification by small protein conjugation or removal [NA]
regulation of post−translational protein modification [NA]
positive regulation of post−translational protein modification [NA]
negative regulation of post−translational protein modification [NA]
protein poly−ADP−ribosylation [NA]
Training_BFR
Training_HLRT
Core genes in child
0
25
50
75
100
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.


proteasomal protein catabolic process
GO:0010498 | core genes: Training_BFR 73, Training_HLRT 79
proteasome−mediated ubiquitin−dependent protein catabolic process [NA]
regulation of proteasomal protein catabolic process [NA]
ERAD pathway [NA]
positive regulation of proteasomal protein catabolic process [NA]
negative regulation of proteasomal protein catabolic process [NA]
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Child test
padj < 0.05
n.s.
Core genes in child
10
20
30
40
50
60


protein modification by small protein conjugation or removal
GO:0070647 | core genes: Training_BFR 89, Training_HLRT 109
protein modification by small protein conjugation [NA]
regulation of protein modification by small protein conjugation or ...
protein modification by small protein removal [NA]
positive regulation of protein modification by small protein conjug...
negative regulation of protein modification by small protein conjug...
Training_BFR
Training_HLRT
Share of parent core genes
1.00
0.75
0.50
0.25
0.00
Core genes in child
25
50
75
100
Child test
padj < 0.05
n.s.